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Canadian Journal of Earth Sciences; Jul 2006; 43, 7; Academic Research Library
pg. 1071

Paleomagnetism of the Wintering Lake pluton and the Early Proterozoic tectonic motion of the Superior Boundary Zone, Manitoba[1,2]

M.J. Harris, D.T.A. Symons, W.H. Blackburn, A. Turek, and D.C. Peck

Abstract: This Lithoprobe-funded paleomagnetic study of the Early Proterozoic Wintering Lake granitoid body supports tectonic models that suggest continental accretion of the Trans-Hudson Orogen with the Superior Craton occurred at ~1822 Ma. Thermal demagnetization data for the granitoid specimens suggest that the magnetic remanence carriers are coarse-grained magnetite or titanomagnetite, and saturation isothermal remanence tests suggest that the magnetite is mostly multidomain. Six of seven paleomagnetic contact tests were negative, indicating that the host rocks have been remagnetized and that the granitoid body may have been partially remagnetized near its margins. Acceptable site mean remanence directions for 20 of 21 granitic sites yield a paleopole at 46.8°N, 102.2°W (with semi-axes of the 95% ellipse of confidence about the paleopole of dp = 11° and dm = 11°). The paleopole fits on the extrapolated apparent polar wander path (APWP) for the Superior craton at ~1822 Ma, which is the interpreted emplacement age of the pluton close to the peak of the Trans-Hudson orogeny. This is the first well-constrained paleomagnetic result from the Superior Province that provides direct evidence from concordant paleopoles for the Early Proterozoic accretion of the orogen to the craton. Further, the paleomagnetic results from the pluton's host rocks, along with other recent results from the Superior Boundary Zone, fill in a gap in the APWP for the craton between ~1780 and ~1720 Ma. The Superior path is now shown to form a hairpin as the craton moves from mid to polar paleolatitudes from ~1880 to ~1830 Ma, suffers a stillstand from ~1830 to ~1770 Ma during the peak of the Trans-Hudson orogeny, returns to mid-paleolatitudes from ~1770 to ~1740 Ma, and then moves on to subequatorial paleolatitudes by ~1720 Ma.

Résumé : Financée par Lithoprobe, cette étude paléomagnétique de l'amas granitoïde de Wintering Lake (Protérozoïque précoce) supporte les modèles tectoniques qui suggèrent que l'accrétion continentale de l'orogène trans-hudsonien et du craton du Supérieur ait eu lieu vers 1822 Ma. Selon les données de démagnétisation thermique prélevées sur des échantillons du granitoïde, les porteurs de rémanence magnétique sont une magnétite ou une titanomagnétite à grains grossiers et, selon des essais de rémanence à saturation isotherme, la magnétite provient de domaines multiples. Six des sept essais de contact paléomagnétique étaient négatifs, ce qui indique que les roches hôtes avaient été remagnétisées et que l'amas granitoïde pourrait avoir été partiellement remagnétisé près des bordures. Des directions moyennes acceptables de rémanence pour 20 de 21 sites granitiques indiquent un paléopôle à 46,8 °N, 102,2 °W (dp = 11°, dm = 11°). Cela concorde bien avec la courbe de dérive apparente extrapolée pour le craton du Supérieur vers 1822 Ma, ce qui serait l'âge de la mise en place du pluton, presque au pic de l'orogène trans-hudsonien. Cela représente le premier résultat paléomagnétique bien encadré de la Province du lac Supérieur qui fournit des preuves directes de paléopôles concordants pour une accrétion de l'orogène au craton au Protérozoïque précoce. De plus, les résultats paléomagnétiques des roches encaissant le pluton, combinés à d'autres résultats récents de la zone limite du Supérieur, fournissent des données manquantes dans la dérive apparente du pôle pour le craton entre ~1780 Ma et ~1720 Ma. La dérive du pôle de la Province du lac Supérieur est maintenant montré sous la forme d'une boucle tournant à angle aigu alors que le craton se déplaçait de paléolatitudes moyennes à polaires de ~1880 Ma à ~1830 Ma, s'est immobilisé de ~1830 Ma à 1770 Ma au sommet de l'orogène trans-hudsonien et est retourné à des paléolatitudes moyennes de ~1770 Ma à ~1740 Ma pour ensuite se déplacer vers

Received 29 April 2005. Accepted 15 February 2006. Published on the NRC Research Press Web site at http://cjes.nrc.ca on 4 September 2006.

Paper handled by Associate Editor F.A. Cook.

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[1]This article is one of a selection of papers published in this Special Issue on *The Western Superior Province Lithoprobe and NATMAP transects.*
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des paléolatitudes subéquatoriales vers ~1720 Ma.

[Traduit par la Rédaction]

Introduction

The Precambrian Shield of North America includes seven Archean geological provinces welded together by Early Proterozoic collision zones. The largest of these collisional orogens, the 1.9–1.7 Ga Trans-Hudson Orogen (THO), separates the northwestern part of the Superior Province in Manitoba from the Rae and Hearne provinces in Saskatchewan and the Northwest Territories (Fig. 1*a*) (Lewry and Collerson 1990). The Lithoprobe THO transect program sponsored 20 paleomagnetic studies (Fig. 1*a*) at the University of Windsor Paleomagnetic Laboratory. These studies have yielded basic data for tectonic modeling of the THO, as recently summarized by Symons (1998), Symons and MacKay (1999), and Symons and Harris (2000, 2005). This database of results from >5900 specimens provides the best paleomagnetic record of any Precambrian collisional orogen yet obtained (M.W. McElhinny, personal communication, 2000).

One of the more significant results from these studies is that apparent polar wander path (APWP) segments in the ~1875 to ~1830 Ma time span for the Flin Flon belt, Hanson Lake block, and Lynn Lake – LaRonge domain all project to convergence at ~1830 Ma (Fig. 2*a*). This is the classical paleomagnetic signature for continental accretion tectonics; however, paleomagnetic data for the 1830–1770 Ma time span, when Trans-Hudson orogenesis came to an end, have only recently been determined.

Results for the ~1810 Ma Wapisu gneiss dome, which is exposed in the eastern part of the Kisseynew domain in central Manitoba (Fig. 1*a*), yielded a paleopole position that defines a sharp ~90° change in trend of apparent polar wander from northeast in today's coordinates between ~1875 and ~1830 Ma to northwest between ~1830 and 1810 Ma (Fig. 2*a*) (Symons and Harris 2000). The observation is important because, as the authors cite, the three most recent hairpins (Irving and Park 1972) in the North American APWP are caused by orogenesis, i.e., the Acadian orogeny at ~375 Ma, the Alleghenian orogeny at ~325 Ma, and the Laramide orogeny beginning at ~130 Ma. Further, the APWP in the THO has been traced from 1795 to 1776 Ma by paleomagnetic results from the Deschambault Lake granites and pegmatites, respectively (Fig. 2*a*) (Symons 1998; Symons et al. 1999, 2000; Symons and Harris 2005).

Tectonic motions for the accretion of the THO to the Superior Craton are not well known. Until recently they have been extrapolated from only two paleopoles, i.e., the 1880 Ma Molson dike swarm (Zhai et al. 1994; Halls and Heaman 2000) and the 1849 Ma Sudbury Complex (Morris 1984) paleopoles (Fig. 2*b*). Four paleomagnetic studies have been undertaken in the Superior Boundary Zone under the aegis of the Lithoprobe Western Superior Transect program. The results of three of these four studies have been published, i.e., for the ~1836 Ma Mystery Lake pluton, the ~1818 Ma Fox Lake pluton, and the ~1770 Ma pegmatites and their contact zones (Harris et al. 1999*a*, 1999*b*, 2000, 2001). This paper presents the results for the fourth study of the ~1822 Ma Wintering Lake pluton and then provides a tectonic synthesis for all four new paleomagnetic results from the Superior Boundary Zone.

Geology

The Wintering Lake pluton intruded supracrustal rocks of the Thompson Nickel Belt about 30 km south of the city of Thompson, Manitoba (Fig. 1*b*). The belt derives its name from its many Ni-sulphide deposits, including the world-class Thompson mine deposit. Most of the nickel deposits are hosted by paragneisses belonging to the Ospwagan Group; however, many of the rocks in the belt are polymetamorphosed migmatitic orthogneisses that probably represent reworked Pikwitonei-type granulites observed in adjacent parts of the Superior Boundary Zone (Weber 1990). Maximum metamorphic conditions for the host rocks of this study range from 600 to 660 °C and 500 to 600 MPa; however, the Wintering Lake pluton is thought to be the product of thermal relaxation and therefore most likely did not see these peak pressure (*P*) – temperature (*T*) conditions (Bleeker 1990).

Around the Wintering Lake pluton (Fig. 3) the host rocks are mafic orthopyroxene granulite, metatonalitic and enderbitic gneiss, migmatitic gneiss, and anatectic gneiss (Hubregtse 1978). Crosscutting amphibolitic mafic dikes of the Molson swarm are typically associated with metatonalitic and enderbitic gneiss. Well-banded migmatitic gneiss occurs within a zone of 0.1–1.0 km from the margin of the pluton.

The Wintering Lake pluton consists dominantly of monzogranite with subordinate amounts of granodiorite and syenogranite. These rocks are massive, medium-grained, and pink to light red. Petrographic observations show that the pluton contains both igneous and metamorphic textures, the latter consisting of recrystallized quartz with embayed or scalloped grain boundaries. The alkali feldspar is dominantly microcline with minor amounts of microperthite, and it is often present as anhedral interstitial megacrysts containing quartz and plagioclase inclusions. Biotite is the main mafic mineral and ranges modally from 2% to 10%. Amphiboles are rare and are severely altered when present. Trace minerals (<2%) include muscovite, apatite, and unidentified opaque minerals. Plagioclase is commonly altered to clay minerals and less commonly to calcite. Biotite is typically altered to chlorite. U–Pb monazite age dating indicates crystallization of the pluton at 1822 ± 3 Ma (Machado et al. 1987). Also, inherited zircons in the pluton have given a U–Pb age of 2590 ± 14 Ma, and the host rocks have given reset U–Pb sphene ages of 1760–1720 Ma (Machado et al. 1987).

Sampled host rocks include tonalitic to granitic gneisses that have biotite as their main mafic mineral. They contain trace amounts of muscovite and unidentified opaque minerals. The amphibolitic gneiss at site 25 is composed dominantly of plagioclase with subordinate (~15%) amphibole and biotite, and with trace amounts of alkali feldspar and unidentified opaque minerals. The mafic dike at sites 05 and 06 consists

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Fig. 1. (*a*) Tectonic elements of the Trans-Hudson Orogen and the location of the Fox Lake pluton (after Harris et al. 1999*a*, 1999*b*, 2000): HLB, Hanson Lake Block; SBZ, Superior Boundary Zone. The solid circles show paleomagnetic collection locations: BB, Baldock batholith (Symons et al. 1996*b*); BP, Boot Lake – Phantom Lake complex (Symons and MacKay 1999); DG, Deschambault Lake granite (Symons et al. 1999); DL, Davin Lake granite (Symons et al. 1996*a*); DP, Deschambault Lake pegmatites (Symons et al. 2000); GL, Granite Lake pluton (abandoned study); HP, Hanson Lake pluton (Gala et al. 1994); JL, Jan Lake granite (Gala et al. 1995); KL, Kississing Lake dome (Symons and Harris 1998); LL, Lynn Lake gabbro (Dunsmore and Symons 1990; Turek et al. 2000); MC, Macoun Lake granite (Symons et al. 1994); MI, Missi conglomerate (Symons and Lewchuk 1994); NM, Namew Lake gabbro (abandoned study); PG, Pelican granulite (abandoned study); PL, Peter Lake gabbro (Symons 1994); RE, Reynard Lake granodiorite (Symons 1995); SG, Sahli granulite (Gala et al. 1998); WB, Wathaman batholith (Symons 1991); WD, Wapisu Lake dome (Symons and Harris 2000). MB, Manitoba; SK, Saskatchewan. (*b*) Locations of the Mystery Lake pluton (Harris et al. 1999*a*), Thompson pegmatites (Harris et al. 2001), and Wintering Lake pluton (this study) collections near Thompson, Manitoba. BRF, Burntwood River fault; CNR, Canadian National Railway. Highways 6, 280, and 391 are shown.

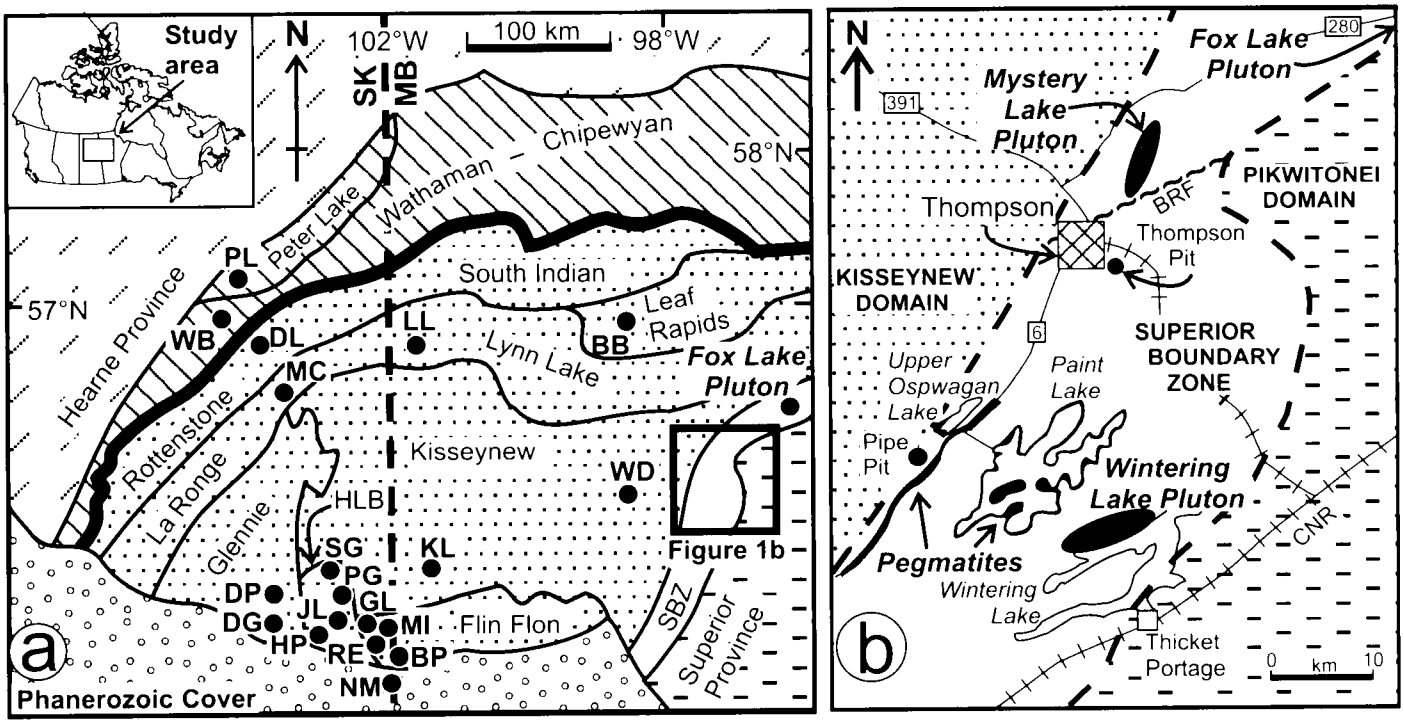
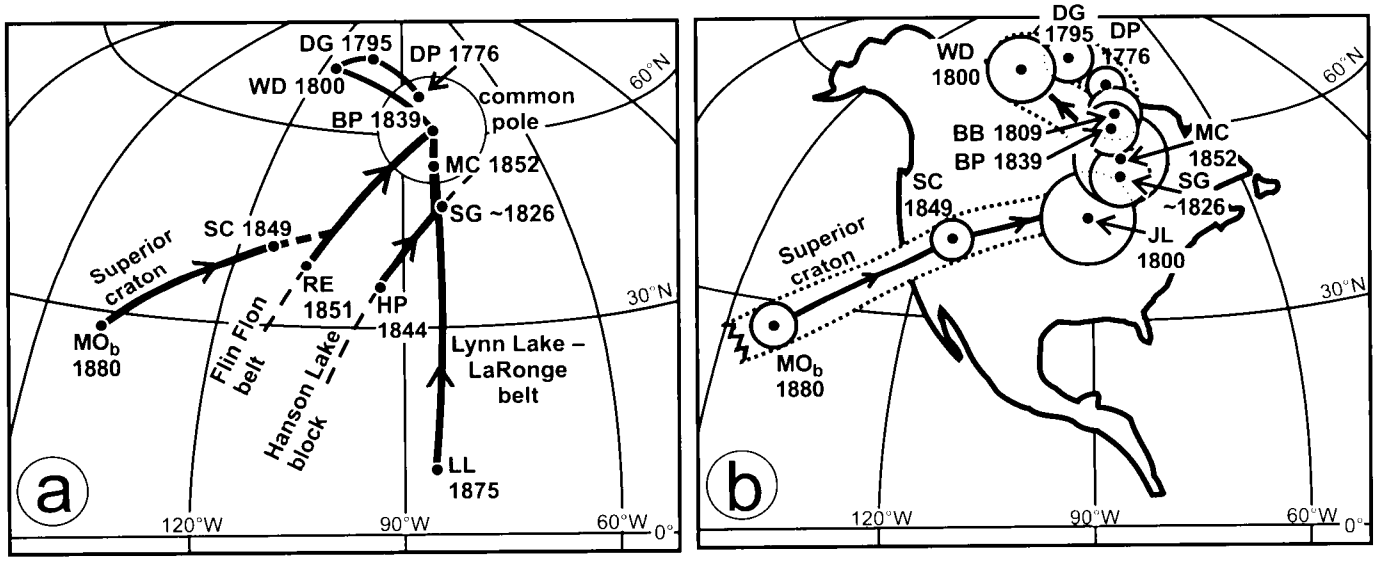


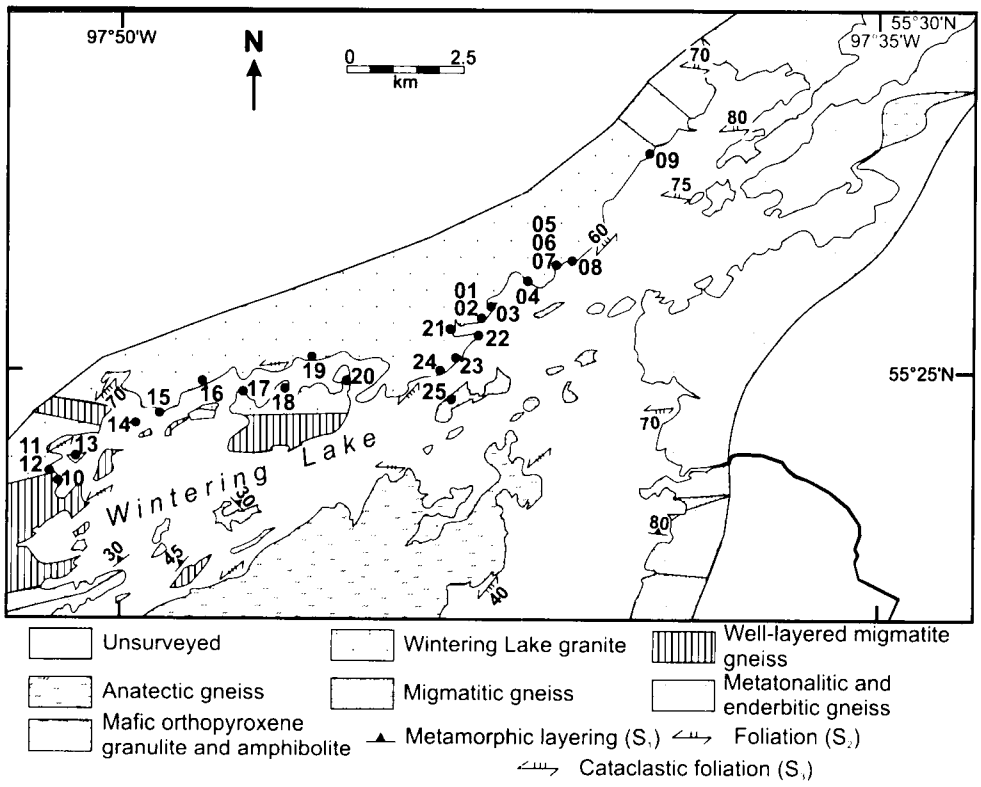
Fig. 2. (*a*) Apparent polar wander path (APWP) segments for the Superior Craton and the Trans-Hudson Orogen domains from ~1880 to ~1776 Ma. (*b*) APWP for the Superior Craton only from 1880 to 1776 Ma (Symons and Harris 2005). Abbreviations for poles as noted in Fig. 1 with their magnetization age in Ma. MO_b, Molson dikes *b* pole (Halls and Heaman 2000); SC, Sudbury complex (Morris 1984).



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Fig. 3. Geology for the Wintering Lake pluton after Hubregtse (1978), with sampling site localities.



mainly of amphibole with minor amounts of plagioclase and trace amounts of olivine and unidentified opaque minerals. The pegmatites collected at sites 10 and 25 have not been found to cut the pluton. They are pinkish-white and massive with a granitic composition and ~10% coarse-grained books of biotite.

Paleomagnetism

Methods

The collection comprises 25 sites of paleomagnetic samples from the northern shore of Wintering Lake (Fig. 3). Twenty-one sites sampled the granitoid, with three of them providing contact tests between the granitoid and the gneisses. Another two sites provided contact tests between the gneiss and a mafic dike, and the remaining two sites provided contact tests between the gneiss and two different pegmatites.

Five to nine cores, with an average of six, were drilled at each site. The cores have a diameter of 2.54 cm and were oriented in situ with a sun compass, with periodic checks using a magnetic compass. A total of 216 specimens, each with a length of 2.2 cm, were sliced from the cores (Table 1).

All paleomagnetic measurements were done at the University of Windsor using an automated CTF DRM-420 cryogenic magnetometer, a Sapphire Instruments SI-2b susceptibility meter, a Sapphire Instruments SI-4 alternating-field demagnetizer, a Sapphire Instruments SI-5 spinner magnetometer, a Sapphire Instruments SI-6 pulse magnetizer, and a Magnetic Measurements MMTD-80 thermal demagnetizer. These instruments are housed in a magnetically shielded room with an ambient field of ~0.2% of the Earth's magnetic field intensity.

The natural remanent magnetization (NRM) was measured for each specimen. The intensity of the specimens from the pluton ranged from 8.1×10^{-5} to 2.7 A/m, with a median intensity of 2.2×10^{-2} A/m, a first quartile (Q_1) of 3.8×10^{-3} A/m, and a third quartile (Q_3) of 4.6×10^{-2} A/m.

Two pilot specimens were selected from each lithology at each site. The chosen specimens had an "average" NRM direction and intensity for the lithology at the site. One pilot specimen was demagnetized in an alternating field (AF) in 11 steps from 5 to 130 mT, and the other was thermally demagnetized in 12 steps from 200 to 570 °C.

In general, the thermal demagnetization intensity curves suggest the presence of titanomagnetite or coarse-grained magnetite in the majority of the specimens, resulting in gradual intensity decay over the entire temperature range of 200–570 °C (Fig. 4, e.g., WL0105, WL0509). A few specimens retained up to 40% of their NRM intensity at 570 °C, suggesting that hematite is present in them (Fig. 4, e.g., WL2004). Only a very few specimens showed pyrrhotite decay curves with distinct intensity drops between 260 and 320 °C.

Evaluation of the thermal decay curves and the orthogonal step demagnetization plots of the AF-demagnetized pilot specimens (Fig. 5) (Zijderveld 1967) suggested that the remaining specimens should be split evenly between thermal and AF demagnetization using a schedule of steps similar to that used for the pilot specimens.

Specimen characteristic remanent magnetization (ChRM) directions were calculated using the end-point method of Kirschvink (1980). Specimen ChRM directions were accepted

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Table 1. Site mean ChRM directions for the Wintering Lake area.

Site[a]	N	n	No. of sites	Dec. (°)	Inc. (°)	α_{95} (°)	k
01Gr	7	7		310.1	88.8	18.3	12
02Gr	8	7		18.3	72.6	15.3	17
03Gr	7	7		309.4	81.4	11.6	28
04Gr	7	7		133.5	82.0	6.6	84
05Gn	7	6		173.6	69.0	7.3	86
05Md	10	10		168.1	64.1	7.1	48
06Gn	7	7		165.1	66.4	6.5	88
06Md	8	8		184.8	65.8	6.2	82
07Gr	2	2		214.3	78.3	15.5	163
07Gn	3	3		145.2	69.4	26.0	24
08Gr	8	8		350.1	67.1	15.4	14
09Gr	6	4		278.2	81.8	24.3	15
10Gn	5	5		105.0	72.8	10.3	57
10Pg	2	2		233.5	74.0	25.2	100
11Gn	7	6		175.1	66.9	3.6	341
11Gr	6	5		180.4	70.2	5.9	171
12Gr	5	5		190.3	76.3	7.5	106
13Gr	6	6		168.0	72.5	12.1	32
14Gr	5	5		178.7	71.6	4.0	370
14Gn	5	5		179.0	70.1	13.5	33
15Gr	7	7		178.5	73.5	5.1	143
16Gr	7	7		195.9	68.5	9.0	45
17Gr	5	5		205.3	71.1	15.2	26
18Gr	9	9		170.3	85.7	9.4	31
19Gr	7	5		0.7	74.9	8.8	77
20Gr	7	6		165.8	83.3	9.0	56
21Gr	13	13		335.6	84.5	10.8	16
22Gr	11	11		290.8	86.1	4.1	124
24Gr	10	10		189.8	87.0	6.4	57
25Pg	5	5		171.5	79.9	20.7	15
25Am	6	5		183.6	79.5	18.5	18
Means							
Gr			20	200.3	85.5	5.5	37
Gn			7	162.0	72.1	7.1	74
Md			2	176.2	65.2	—	250
Pg			2	210.1	78.7	—	64
Gn + Md + Pg			11	170.8	72.6	5.9	61

Note: Dec., declination; Inc., inclination; k, precision parameter (Fisher 1953); N, total number of specimens; n, number of specimens used in the averages; α_{95}, radius of the cone of 95% confidence.
[a]Site number and abbreviation for rock type: Am, amphibolite; Gn, granitic gneiss; Gr, granite; Md, mafic dike; Pg, pegmatite.

for statistical analysis if their best-fit stable direction had a maximum deviation of <20° over three or more consecutive steps, with most having deviations of <10°. Site and unit mean directions were calculated using the statistical methods of Fisher (1953).

The majority of specimens possessed a single ChRM component that decayed linearly to the origin in orthogonal step demagnetization diagrams (Fig. 5*b*). Some specimens carried a weak viscous remanence in the present-day Earth's magnetic field direction, but this was removed by the initial demagnetization steps (Fig. 5*a*). A few specimens contained a minor amount of hematite and thus were not completely demagnetized by either AF or thermal techniques (Fig. 5*c*).

Results

Acceptable mean directions were calculated for all sites using the majority of the specimens (Table 1), except for site 23, which gave very scattered directions. All of the 24 accepted sites yielded mostly normal-polarity, steeply down inclinations (Fig. 6*a*). However, about 15% of the specimens,

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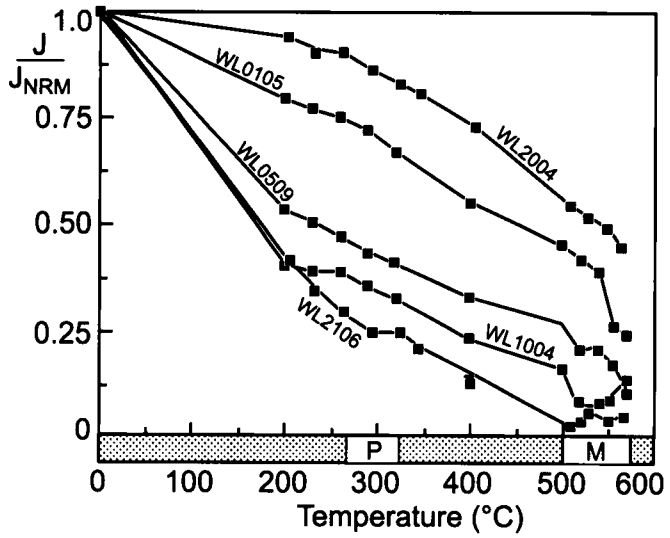


Fig. 4. Thermal decay curves for representative specimens of the Wintering Lake pluton. Curve labels denote the collection, site (s), and core (c) numbers (e.g., WLsscc). Typical unblocking temperature ranges for pyrrhotite (P) and magnetite (M) are shown along the bottom. J/J_{NRM} is the ratio of the measured intensity to the NRM intensity.

representing several sites in part, yielded reverse-polarity ChRM directions. About half of these reverse-polarity specimens provided antipodal directions to the normal-polarity specimens, and so these antipodal directions have been inverted to normal polarity to calculate the site mean direction.

The 20 site mean directions for the Wintering Lake pluton (Gr in Table 1) combined to give a unit mean direction of declination (Dec.) of 200.3°, an inclination (Inc.) of 85.5°, a radius of the cone of 95% confidence (α_{95}) of 5.5°, and a precision parameter (k) of 37. Although there is a low error on the mean direction, the population is elongated north–south (Fig. 6*a*). This may imply that the pluton cooled relatively slowly, thus acquiring its magnetization over a few million years as the Earth's pole migrated. When the α_{95} errors associated with each site are considered (Fig. 6*a*), however, even the three most northerly sites (02, 08, and 19) that appear most isolated cannot be separated statistically from adjacent site means in the population. Nor are there grounds to isolate this group on either geological or geographical data. Lastly, these three site mean directions are <3 SD (standard deviations) from the overall mean of all granitic pluton sites. Therefore, even these three sites do not form a separate population, and thus there is only one statistical population of granitic site mean directions.

The seven host rock sites of granitic gneiss yielded a unit mean direction of Dec. = 162.0°, Inc. = 72.1°, α_{95} = 7.1°, and k = 74; the two mafic dike sites gave Dec. = 176.2°, Inc. = 65.2°, and k = 250; and the two pegmatite sites gave Dec. = 210.1°, Inc. = 78.7°, and k = 64 (Table 1; Fig. 6*b*).

The mean directions for the gneisses and the mafic dikes were compared using the F test of McFadden and Lowes (1981) to determine if the two clusters could be drawn from a single population or represent separate populations. The test proved negative because $F_{calc} = 1.4 \ll F_{stat} = 5.8$ (total number of site mean directions N = 9); thus the two clusters can be drawn from the same population, i.e., the gneisses and mafic dikes may carry the same magnetization. Although the site mean directions for the pegmatites are statistically poorly defined, i.e., $\alpha_{95} > 20°$, the F test was used to compare their directions with those of the gneisses and dikes. The test also was negative with $F_{calc} = 3.4 < F_{stat} = 4.7$ (N = 11). Note that the age relations between the pluton and the dikes or pegmatites cannot be readily determined because neither the dikes nor pegmatites were seen in contact with the granite. The combined mean direction for the host rock, dike, and pegmatite sites is Dec. = 170.8°, Inc. = 72.6°, α_{95} = 5.9°, and k = 61 (Table 1; Fig. 6*c*). Lastly, when the plutonic site mean directions were compared with the combined direction for the gneisses, dikes, and pegmatites, the test is positive, with $F_{calc} = 8.3 \gg F_{stat} = 3.3$ (N = 27). This suggests that the pluton carries a different magnetization from that of the other three lithologies (Fig. 6*c*).

Contact tests

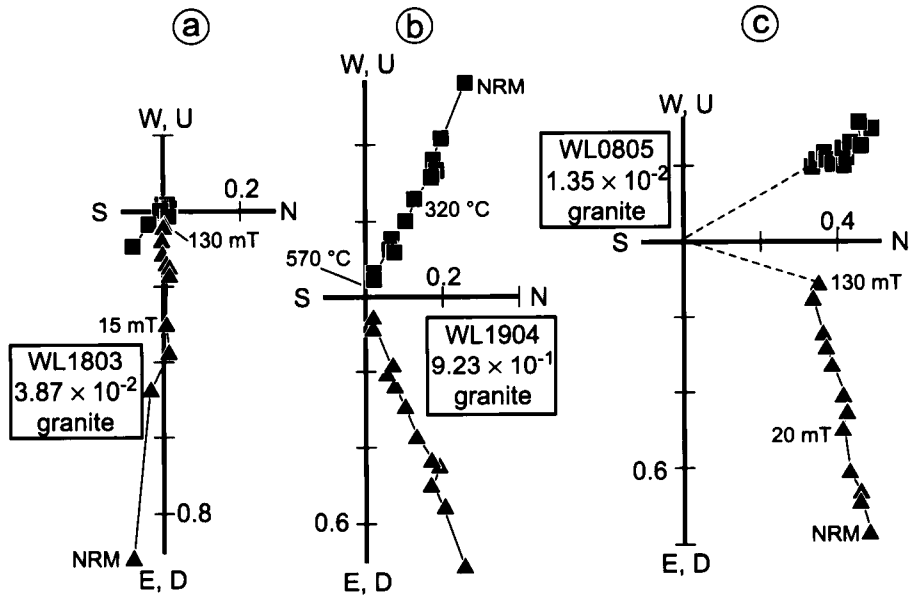
Seven sites were collected as paleomagnetic contact tests, sampling two different lithologies at each site. If the mean directions for the two lithologies are similar, then it implies that they carry the same magnetization component. If the directions differ, then the magnetization events differ and their relative timing may be evaluated. Two of the contact tests were between the gneisses and the mafic dike at sites 05 and 06 (Fig. 3; Table 1), and both tests were negative, i.e., $F_{calc} = 1.0 \ll F_{stat} = 3.7$ (total number of specimens n = 16) and $F_{calc} = 2.9 < F_{stat} = 3.8$ (n = 15), respectively, suggesting that both lithologies record the same magnetization direction. At site 25 between a pegmatite and amphibolitic gneiss, the test was also negative, with the directions being so similar that an F test was not needed. At site 10, a contact test between a pegmatite and granitic gneiss was positive, with $F_{calc} = 11.8 > F_{stat} = 5.8$ (n = 7); however, the pegmatite population comprised only two specimens and thus had a high α_{95} value, indicating caution using this test. Thus three of these four contact tests indicate that the granitic and amphibolitic gneisses, pegmatites, and mafic dikes carry the same ChRM direction. These results, along with the similar mean directions for the three units reported in the preceding section, indicate that all of the host rocks were magnetized or remagnetized at the same time.

Contact tests were carried out at three sites (07, 11, 14) between the Wintering Lake pluton and granitic gneiss host rocks. All three tests in three widely separated outcrops were negative, with the granitic gneiss giving the same ChRM direction statistically as the abutting pluton. Recall from the previous section that the pluton and its host rocks regionally give statistically different paleopole directions at 95% confidence. Thus, these contact tests indicate that the host rocks were remagnetized next to their contact with the Wintering Lake pluton contact, whereas farther away they were not remagnetized to the same ChRM direction as the pluton. On a regional basis, therefore, the pluton yields a positive contact test. This result suggests that the remanence of the pluton is primary. Combined with the negative contact tests between the host rock units, one of two scenarios is most likely. The first scenario is that the pluton was emplaced with a baked zone into host rocks with an older remanence from regional metamorphism, which is the classic interpretation of a positive

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Fig. 5. Representative orthogonal step demagnetization plots (Zijderveld 1967) for specimens from the Wintering Lake pluton. Squares are projected in the horizontal plane (west (W) – north (N) – east (E) – south (S)) and triangles are projected in the vertical plane (up (U) – north (N) – down (D) – south (S)). Label indicates collection, site, and core as in Fig. 4; NRM intensity in A/m; and lithology.



contact test. The other scenario is that the host rocks were preferentially remagnetized by hydrothermal fluids after the pluton was emplaced, whereas the highly impermeable pluton was not remagnetized. In the latter scenario, the common direction in the contact zone could be caused by either the "baked" zone rocks preferentially resisting remagnetization from physiochemical changes upon the pluton's intrusion, or by partial alteration of the pluton's margin by the hydrothermal fluids.

Discussion of paleopoles

The 1880–1776 Ma Superior Craton polar wander path

The paleopoles obtained from the Superior Boundary Zone may be compared with the APWP for the Superior Craton only prior to amalgamation with the THO and with the APWP for both the craton and orogen upon and after accretion, as discussed extensively in Symons and Harris (2005). Excluding data from the Superior Boundary Zone, this pre-accretion path is defined by only the 1880 Ma Molson dike *b*-component paleopole (MO_b) (Halls and Heaman 2000) and the 1849 Ma Sudbury Complex paleopole (SC, Fig. 2*a*) (Morris 1984). Accretion of the orogen to the craton is defined by the intersection of the three terrane APWP segments that track through the paleopoles of the 1852 ± 4 Ma Macoun Lake granite (MC) (Symons et al. 1994), the 1839 ± 3 Ma Boot–Phantom pluton (BP) (Symons and MacKay 1999), and the 1826 ± 10 Ma Sahli charnockite (SG) (Gala et al. 1998). Lastly, five paleopoles define the post-accretion APWP for the combined craton. They come from the 1800 ± 15 Ma Wapisu gneiss (WD) (Symons and Harris 2000), the 1800 ± 14 Ma Jan Lake complex (JL) (Gala et al. 1995), the 1809 ± 6 Ma Baldock batholith (BB) (Symons et al. 1996*b*), the 1795 ± 15 Ma Deschambault Post plutons (DG) (Symons et al. 1999), and the 1776 ± 5 Ma Deschambault pegmatites (DP, Fig. 2*b*) (Symons et al. 2000). The assignment of ages to the paleopoles is based on the available direct or bracketing radiometric age dates for the rock units along with either the reported error limits of single determinations or the root mean square errors of multiple determinations as given in the original paper or updated in Symons and Harris (2005).

Mystery Lake pluton

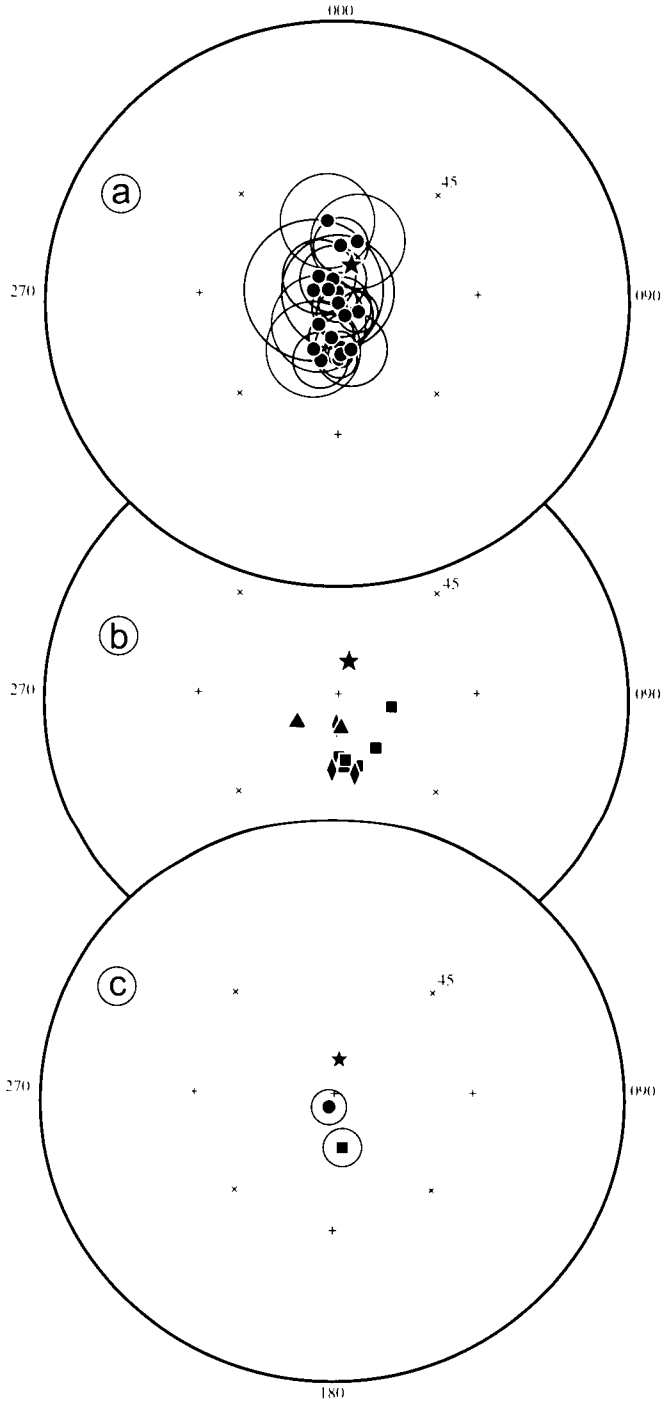
Paleomagnetic results have been reported from the Mystery Lake pluton, which is centred about 15 km north of the town of Thompson, Manitoba, in the Thompson Nickel Belt (Fig. 1*b*) (Harris et al. 1999*a*). This granitoid pluton is about 25 km long, trending ~N20°E, and about 4 km wide. It intrudes Early Proterozoic pelitic metasedimentary and mafic metavolcanic rocks of the Ospwagan Group (Weber and Scoates 1976). The pluton is composed mainly of granodiorite and tonalite and has given a $^{207}Pb/^{206}Pb$ monazite age of 1836 Ma (Syme et al. 1993) that has widely been interpreted as the age of intrusion rather than as a minimum age. The northern end of the pluton is prominently foliated, whereas the southern end is massive to weakly contact foliated. The contact foliations dip steeply towards the margins in the pluton and steeply away from the margins in the host rocks, which implies a vertical emplacement of the pluton without subsequent tilting. Also, the pluton is cut by northeast- and northwest-trending mafic dikes, with the latter known to belong to the ~1250 Ma Mackenzie dike swarm.

Eleven of the 14 sites in the foliated granitoids from the northern part of the pluton gave reliable site mean ChRM directions, with a unit mean direction of Dec. = 254.9°, Inc. 74.2°, α_{95} = 10.5°, and k = 22 that gives a pole position of 40°N, 136°W (semi-axes of the 95% ellipse of confidence about the paleopole of dp = 17° and dm = 19°; Fig. 7). This pole position falls off the APWP for the Superior Craton, being closest to the ~1865 Ma part of the path, which in turn suggests that the 1836 Ma age from $^{207}Pb/^{206}Pb$ monazite is an unreliable minimum age. This finding was reinforced by the paleomagnetic results from the one northwest-trending

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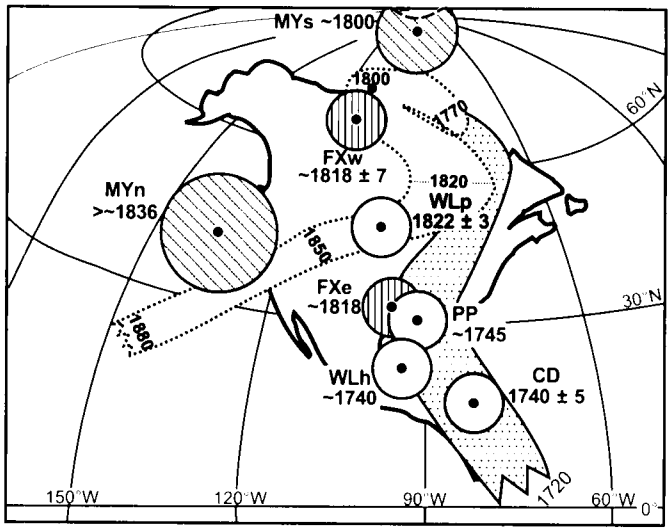
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Fig. 6. Equal-area lower-hemisphere projection showing (*a*) site mean directions (●) for the Wintering Lake pluton granitoids; (*b*) site mean directions for the granitic gneiss (■), pegmatite (▲), and metamorphosed dikes and amphibolite (◆); and (*c*) mean directions for the plutonic (●) and host rock (■) sites. The cones of 95% confidence are shown in (*a*) and (*c*). ★, present Earth's magnetic field direction.



mafic dike that was observed and sampled (site 6) (Harris et al. 1999*a*) because it gave a virtual pole position that is close to the 1880 Ma (MO_b) paleopole (Halls and Heaman 2000), which required that the foliated northern part of the

Fig. 7. Paleopoles for collections from the Superior Craton from about 1880–1720 Ma with the addition of paleopoles from the Cleaver dikes (CD) (Irving et al. 2004) and the Superior Boundary Zone (Fox Lake pluton eastern Split Lake area (FXe) and western Little Assean Lake area (FXw) (Harris et al. 2000)), Mystery Lake pluton foliated northern (MYn) and southern massive (MYs) areas (Harris et al. 1999*a*), Pipe mine – Paint Lake pegmatites (PP) (Harris et al. 2001), and Wintering Lake pluton (WLp) and host rocks (WLh; this study). Abbreviations as in Figs. 1 and 2.



Mystery Lake pluton pre-date 1880 Ma. Prior to this finding, Molson dikes had not been reported cutting the Mystery Lake pluton. Since then, Zwanzig (2000) has mapped these foliated rocks as basement gneisses and not deformed granite, and the presence of a Molson dike intruding the gneisses has been confirmed (D.C. Peck, personal communication, 2004). One possible older age for the granitic gneisses is ~2140 Ma from comparison with diabase dike paleopoles (Buchan et al. 1996; Harris et al. 2001), and the other possible older age is ~2275 Ma from comparison with the APWP for the Superior Craton given by Dunsmore and Symons (1990). It is also noteworthy that Harris et al. (1999*a*) obtained a positive contact test between the granite gneisses and the host rock metasediments, implying different ages of remanence acquisition.

All four sites collected in the massive to slightly foliated southern portion of the Mystery Lake pluton gave site mean ChRM directions with a mean of Dec. = 359.2°, Inc. = 70.7°, α_{95} = 13.5°, and k = 47, yielding a paleopole at 89°N, 109°W (dp = 20°, dm = 23°; Fig. 7). Noting that the region between the exposed northern granitic gneisses and southern massive granodiorite is covered, it is possible that the southern portion is a younger pluton emplaced into the same zone of structural weakness as that of the northern portion. In this scenario, the likely age of the younger southern pluton is ~1800 Ma, as the paleopole abuts the APWP for the Superior Craton at this time. Alternatively, if the southern granodiorite is an unmetamorphosed portion of the original pluton that had its northern portion subsequently metamorphosed and presumably remagnetized, then the remanence of the southern portion should pre-date the suggested ages of ~2140 or ~2275 Ma for the northern gneisses. In this alter-

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native scenario, the likely age for the massive southern portion based on its paleopole is ~2500 Ma using the APWP for the Superior Craton of Vandall and Symons (1990).

Geochronological investigations are ongoing to help resolve the crystallization and metamorphic ages of the Mystery Lake pluton but have so far proven inconclusive (e.g., Peck 1999; Zwanzig 2000; Harris et al. 2001).

Lastly, one of the northwest-trending mafic dikes that intrudes the foliated granitic gneisses yields a paleopole that is similar to those from the ~1270 Ma Mackenzie diabase dike swarm (site 12) (fig. 8-8 in Harris et al. 1999*a*), confirming that the dike belongs to the Mackenzie dike swarm (ref. Weber 1976; Bleeker 1990).

Fox Lake pluton

The second paleomagnetic study was on the Fox Lake pluton, located about 100 km northeast of Thompson in the Split Lake Block of the Superior Boundary Zone (Fig. 1*a*) (Harris et al. 2000). This pluton outcrops over an area of >400 km^2, but outcrop is sparse, being limited to the shores of Little Assean Lake and Awupak Narrows, where the pluton is composed of a highly foliated biotite granodiorite (14 sites), and along the shores of Split Lake, where the pluton is massive to slightly foliated biotite granodiorite (six sites). The pluton is intruded mostly into hornblende–biotite gneiss that has yielded Archean $^{207}Pb/^{206}Pb$ zircon ages between 2822 and 2617 Ma (Böhm et al. 1999) as well as into minor amphibolite. The pluton itself has given $^{207}Pb/^{206}Pb$ titanite model ages of 1811 Ma (Heaman and Corkery 1996) and 1825 ± 6 Ma (L.M. Heaman, personal communication, 2000), suggesting a minimum age for emplacement of ~1818 ± 7 Ma. In addition, Lowdon (1961) determined a K–Ar biotite age of 1720 Ma that could be recording a later deformation or metamorphic event. Four of five paleomagnetic contact tests gave useful results. The one positive test was between weakly foliated granodiorite of the pluton on Split Lake and granitic gneiss of the host rock, indicating different magnetization ages. A negative test was obtained at site 10 (Harris et al. 2000), where a weakly deformed mafic dike, the pluton, and granitic gneiss host rock all gave statistically similar directions, indicating similar ages of magnetization. Negative contact tests were also obtained between the weakly foliated pluton and deformed amphibolitic gneiss host rock on Split Lake, and between the highly foliated pluton and a highly deformed dike on Little Assean Lake.

Paleomagnetic analysis shows two populations of directions. The samples from the six Split Lake sites retain a unit mean ChRM direction of Dec. = 177°, Inc. = 77°, α_{95} = 10.7°, and k = 40, giving a paleopole (FXe, Fig. 7) at 31°N, 95°W (dp = 19°, dm = 20°). This paleopole is close to several paleopoles from the Superior Craton and adjacent THO (Fig. 7) that are dated at 1844, 1849, and 1851 Ma, suggesting an emplacement age of ~1848 ± 4 Ma that is consistent with the $^{207}Pb/^{206}Pb$ titanite ages that suggest an age of ~1818 ± 7 Ma, as noted in the previous paragraph.

In contrast, the samples from the accepted eight sites on Little Assean Lake retain a unit mean ChRM direction of Dec. = 330°, Inc. = 83°, α_{95} = 11.3°, and k = 25, giving a paleopole (FXw) at 67°N, 114°W (dp = 22°, dm = 22°). Again using the F test of McFadden and Lowes (1981) to compare the Little Assean Lake and Split Lake populations of site mean directions gives F_{calc} = 11.75 » F_{stat} = 3.55, which shows that the two populations are significantly different at 95% confidence. This paleopole falls on the APWP for the Superior Craton at the time of collision and accretion of the THO. Its location on the APWP coincides in time with the $^{207}Pb/^{206}Pb$ titanite age of ~1818 ± 7 Ma for the pluton and with the elevated metamorphic temperatures expected to be associated with the adjacent mylonitic shear zone along the Little Assean Lake – Awupack Narrows lineament.

Thompson pegmatites and host rock gneisses

The third study done in the Superior Boundary Zone targeted the ~1770 Ma pegmatites and their contact zones within the host gneisses (Harris et al. 2001). The initial collection of 16 pegmatites came from the INCO Ltd. past-producing Pipe open pit mine (eight sites) and presently producing Thompson open pit mine (eight sites) in the Thompson Nickel Belt. Of these sites, only two of the pegmatite sites from the Pipe mine yielded useful site mean ChRM directions. It was noticed, however, that the contact zones of the pegmatites with their host rocks in the Pipe mine also retained a population of consistent ChRM directions. Therefore, the ChRM directions for these specimens were grouped by lithology (granite gneiss, metabasalt, and mafic dikes) and the mean directions for each calculated as a "site." The modest success of this initial campaign led to a second campaign in which an additional 16 sites were established, with both pegmatite and its host rock contact zone being collected at each site. The sites were located along the shores of Upper Ospwagan Lake (five sites) and Paint Lake (seven sites) and along Highway 6 (four sites).

The supracrustal host rocks in the study area are polymetamorphosed migmatitic orthogneisses, probably reworked Pikwitonei-type granulite (Weber 1990), formed at temperatures of 600–660 °C and pressures of 500–600 MPa (Bleeker 1990). Sampled host rocks giving useful site mean ChRM directions in the second campaign included the granitic gneiss (10 sites), amphibolite (two sites), metabasalt (one site), and a mafic dike (one site). The post-tectonic pegmatites are massive, medium to coarse grained, and leucocratic with less than 5% biotite. Stable coherent site mean ChRM directions were isolated in seven of the 16 pegmatites.

The paleomagnetic results for both sampling campaigns are discussed in detail by Harris et al. (2001). In brief, the mean ChRM direction from the nine pegmatite sites was Dec. = 152.3°, Inc. = 79.1°, α_{95} = 11.6°, and k = 21, which is statistically indistinguishable at 95% confidence from the mean directions from the 13 granitic gneiss sites of Dec. = 199.2°, Inc. = 75.6°, α_{95} = 14.8°, and k = 9 and three amphibolite and mafic dike sites of Dec. = 188.1°, Inc. 68.0°, α_{95} = 17.6°, and k = 50. The latter population excludes aberrant directions for mafic dikes at sites 07 in the Pipe pit and 05 in Upper Ospwagan Lake. One aberrant site mean ChRM direction from granitic gneiss at site 12 in Paint Lake is also excluded. Combining the pegmatite and contact-zone granitic gneiss sites into one population yields a paleopole (PP) at 30.7°N, 92.2°W (dp = 10°, dm = 11°, Fig. 7). The interpretation of these paleomagnetic results from the Pipe mine to the Paint Lake area is discussed in the next section with those from the adjacent Wintering Lake pluton (Fig. 1*b*).

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Wintering Lake pluton

The mean remanence direction for the 1822 ± 3 Ma Wintering Lake pluton yields a paleopole (WLp) at 46.8°N, 102.2°W (dp = 11°, dm = 11°, Fig. 7). This location fits on the APWP for the Superior craton determined by extrapolating the 1880–1849 Ma APWP segment to an ~1822 Ma position (Figs. 2*b*, 7). Note in Fig. 2*b* that five paleopoles (BB, BP, JL, MC, and SG) on the APWP have a mean age of ~1825 Ma. Therefore, the WLp paleopole is interpreted to be a primary thermoremanent magnetization that was acquired during crystallization of the pluton at 1822 ± 3 Ma. This interpretation also considers that the secondary alteration seen in the pluton of the trace amphiboles, minor biotite, and some feldspars has not been sufficiently intense to overwhelm and significantly realign the remanence retained by the unaltered primary magnetic minerals in the granite.

The location of the paleopole (WLh) for the host rock gneisses, pegmatites, and dikes is located at 23.6°N, 93.3°W (dp = 9°, dm = 11°, Fig. 7). As noted earlier, it is not possible to establish the magmatic age relationships between the Wintering Lake pluton and the dikes and pegmatites. Therefore, with the overall *F* test proving negative, the magnetization of the gneisses, dikes, and pegmatites could be older or younger than that of the pluton. Three lines of reasoning suggest that the magnetization is younger than the pluton. First, but obviously not definitive, dated pegmatites that are older than the 1822 Ma pluton are rare (Syme et al. 1993); however, there are many in the 1780–1760 Ma age range, particularly in the Thompson Nickel Belt. Second, because the host rocks were sedimentary (Weber 1990), they would have been far more permeable to fluid flow, transmissive to heat, and ductile upon deformation than the Wintering Lake pluton. Thus it is possible that the sediments were significantly remagnetized during a 1770 Ma or post-1770 Ma metamorphic event, whereas only the margins of the pluton were partially remagnetized. Third, the location of the paleopole for the gneisses, dikes, and pegmatites coincides with the paleopole (PP) from the gneisses and pegmatites in the Pipe Pit – Paint Lake area described earlier (Harris et al. 2001). Many of those pegmatites are confidently dated to be part of the ~1770 Ma pegmatite event in the Thompson Nickel Belt (Syme et al. 1993). These facts suggest that the magnetization of the pegmatites and the remagnetization of the gneisses happened at ~1770 Ma or later but, more importantly, well after the emplacement of the pluton.

Since the WLh paleopole includes sites in post-tectonic pegmatites, its magnetization can be no older than the age of the pegmatites. Unfortunately, there is a range of ages for them. A pegmatite from just northwest of Paint Lake yielded a U–Pb zircon age of 1786^{+3}_{-2} Ma (Krogh et al. 1985), and another from the Pipe Pit yielded a Pb–Pb zircon age of 1768 Ma (Machado et al. 1987). There is general consensus that abundant pegmatite emplacement occurred between 1780 and 1760 Ma; however, the pegmatites have also given several uplift or reset ages at about 1720 Ma (Machado 1990; Weber 1990). Therefore, it is suggested that the Pipe Pit – Paint Lake – Wintering Lake area pegmatites may carry a magnetization that is between ~1770 and ~1720 Ma, i.e., ca. 1745 Ma.

Further, approximately 10 km south of Wintering Lake is the Sasagiu Rapids granite pluton, which has yielded a U–Pb zircon age of 1726 ± 12 Ma that is interpreted to be the age of intrusion (Machado et al. 1990). There are also several radiometric ages between 1730 and 1720 Ma that are interpreted to be metamorphic or reset ages for various gneisses in the Sasagiu area (Lowdon 1961; Lowdon et al. 1963; Machado et al. 1990). The data imply that the pegmatites may carry a primary magnetization acquired between 1786 and 1768 Ma or a subsequent remagnetization between 1768 and ~1720 Ma, and that remagnetization of the amphibolites and gneisses also probably occurred during the same time interval (Harris et al. 2001).

Unfortunately, there appears to be only one reliably dated paleopole from the assembled Superior – Trans-Hudson – Hearne – Rae – Slave craton between ~1770 and ~1720 Ma. It comes from the distant 1740 ± 5 Ma Cleaver dikes (CD) on the shores of Great Bear Lake and is located at 19.4°N, 83.3°W (A_{95} = 6.1°, Fig. 7) (Irving et al. 2004). The location of this 1740 Ma CD paleopole suggests an age of ~1740 Ma for the WLh magnetization of the host rocks and pegmatites of the Wintering Lake pluton and the adjacent region to the west in the Thompson Nickel Belt. The paleopoles also imply that the APWP for the amalgamated craton tracks southward towards lower latitudes between ~1770 and ~1740 Ma, as previously suggested by Halls and Heaman (2000) in their discussion of the Molson dikes A-paleopole and by Harris et al. (2001) in their initial discussion of the pegmatites of the Thompson Nickel Belt.

Also from the average of two primary plutonic poles of ~1750 ± 10 Ma, including the Cleaver dikes pole, and 14 post-Huronian fluid flow and uplift remagnetization poles of ~1710 ± 60 Ma, Irving et al. (2004) obtained a similar southerly pole position at 20.8°N, 94.5°W (A_{95} = 5.2°) for the Slave–Rae–Hearne portion of the Laurentian Shield.

The fact that the host rocks analyzed in this sequence of paleomagnetic studies, as well as the metamorphic parts of the Mystery Lake and Fox Lake plutons, give paleopoles of different ages is not surprising. This is because recent U–Pb zircon ages for metamorphism in the Thompson Nickel Belt indicate a protracted metamorphic history yielding ages that span about 130 Ma, from ~1850 to ~1720 Ma (N. Machado, personal communication, 2001). This suggests that these host rocks have been metamorphosed by both regional and local events at different times in differing locations. Thus the ages interpreted from the available radiometric ages and paleopoles for the plutons and host rocks in this study are plausible initial estimates that await radiometric confirmation or modification.

Tectonic motions

Comparison of the APWP segments for the Superior Craton and THO from ~1880 to ~1820 Ma indicates that the collision and accretion process was one of sinistral transpression as the intervening Manikewan Ocean closed. This sense of motion agrees with the geologic field relationships (Lewry 1981; Bleeker 1990) and plate tectonic modeling (Gibb 1983; Green et al. 1985; Hoffman 1988) in which the Thompson Nickel Belt is an early collisional zone dominated by north–south to northwest–southeast oblique convergence. During this ~1880 to ~1820 Ma period, the Superior Craton was moving at a rate of ~0.8°/Ma or ~8 cm/a, the THO was also moving at ~0.8°/Ma or 8 cm/a, and the Superior Craton and

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THO were closing together at a rate of ~0.55°/Ma (5.5 cm/a) or at an angle of about 40°.

During the interval from ~1810 to ~1770 Ma, the assembled craton was in a stillstand, during which the APWP described a clockwise loop with a diameter of ~10°. This loop is remarkably similar to the counterclockwise loop of ~10° observed in the North American APWP during the Laramide orogenic stillstand from ~120 to 60 Ma (Besse and Courtillot 2002).

Lastly, from ~1770 to ~1740 Ma, the rate of apparent polar wander again increased to a relatively rapid rate of ~1.5°/Ma or 15 cm/a as the entire craton moved from a subpolar to a subequatorial paleolatitude. Although fast, even faster motions of large continents such as Laurentia, Gondawana, and India have been well documented for comparable 30 Ma time intervals during the Phanerozoic (Meert et al. 1993).

Conclusions

Paleomagnetic results from the ~1822 Ma Wintering Lake pluton yield a paleopole at 46.8°N, 102.2°W (dp = 11°, dm = 11°), which fits the extrapolation of the Superior Craton apparent polar wander path (APWP) from 1880 through 1849 to 1822 Ma, about the time of continental accretion with the Trans-Hudson Orogen (THO). Pegmatites, gneisses, and amphibolites (the host rocks for the pluton) appear to carry a remagnetization that yields an ~1745 Ma paleopole at 23.6°N, 93.3°W. It and other paleopoles from the Superior Boundary Zone extend the APWP for the craton from the 1776 Ma Deschambault Lake pegmatite paleopole down to ~1720 Ma. Thus, the APWP of the craton forms a hairpin. It shows that the craton moved from mid-paleolatitudes at ~1880 Ma to polar paleolatitudes by ~1830 Ma, stalled in a stillstand from ~1830 to ~1770 Ma during the peak of the Trans-Hudson orogeny, and tracked back to mid-paleolatitudes by ~1740 Ma and on to subequatorial paleolatitudes by ~1720 Ma. Clearly, it would be highly desirable to have much better age dating constraints in the rock units examined in this series of paleomagnetic studies, something the authors attempted (Harris et al. 1999*a*, 1999*b*) without success, to more precisely outline the tectonic history of the Superior Boundary Zone.

Acknowledgments

The authors wish to thank P.J.A. McCausland for field assistance, Shelie Ann Cascadden for laboratory assistance, Neill Brandon and the Manitoba Department of Energy and Mines for logistic support, the referees M.E. Evans and H.C. Halls and the Associate Editor F.A. Cook for very helpful comments on the manuscript, and the Lithoprobe Western Superior Transect for financial assistance to the Wintering Lake paleomagnetic study.

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